Wednesday, September 10, 2025

LEC4 - The Probability Set Function Continued

Sample Point Method

1. Define the experiment and clearly determine how to describe one simple event
2. List the simple events and test to make sure they cannot be decomposed (Defines S)
3. Assign reasonable probabilities to the sample points in S, making sure $P(A_i) \geq 0$ and $P(\cup_i A_i) = 1$
4. Define the event of interest A as a specific collection of sample points
5. Find P(A) by summing the probabilities of the sample points in A

Counting Sample Points

- A sample space N contains N equally likely events. An event A contains n_a sample points, then $P(A) = \frac{n_a}{N}$
    - If we know N and can count n_a, then we can calculate P(A)

Example

Consider a fair six-sided die. Let A_i $(i = 1, 2, \cdots, 6)$ denote the event "Observe number i".

Then, by axioms 2 and 3:

$$P(S) = P(A_1 \cup A_2 \cup \cdots \cup A_6)$$
$$= P(A_1) + P(A_2) + \cdots + P(A_6) = 1$$

Pros and Cons of Sample-Point Method

- Direct and powerful, but prone to human error
- Mistakes include mis-defining simple events and missing simple events. For sample spaces with a large number of events, can be difficult (or impossible) to count all events
- Leads us to learn about combinatorial analysis (combinations and permutations)

Claim: $P(\emptyset) = 0$

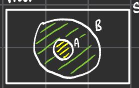
Proof:

$\emptyset \cap S = \emptyset$, $P(S) = 1 = P(S \cup \emptyset) = P(S) + P(\emptyset)$

Therefore, $P(\emptyset) = 0$ as required

Claim: $A \subset B \Rightarrow P(A) \leq P(B)$

Proof:



$B = A \cup (B \cap A^c)$

$P(B) = P(A \cup (B \cap A^c)) = P(A) + P(B \cap A^c)$

By axiom 1, $P(x) > 0$, so P(B) is the sum of 2 positive numbers, so $P(B) \geq P(A)$, as required

Claim: $P(A) \leq 1$

Proof:

$A \subset S$, so $P(A) \leq P(S) = 1$

Therefore $P(A) \leq 1$

Claim: $P(A^c) = 1 - P(A)$

Proof:

$A \cup A^c = S$, where $A \cap A^c = \emptyset$ (Mutually exclusive)

$P(A \cup A^c) = S = 1$, by axiom 2

$P(A) + P(A^c) = 1$

Therefore, $P(A^c) = 1 - P(A)$